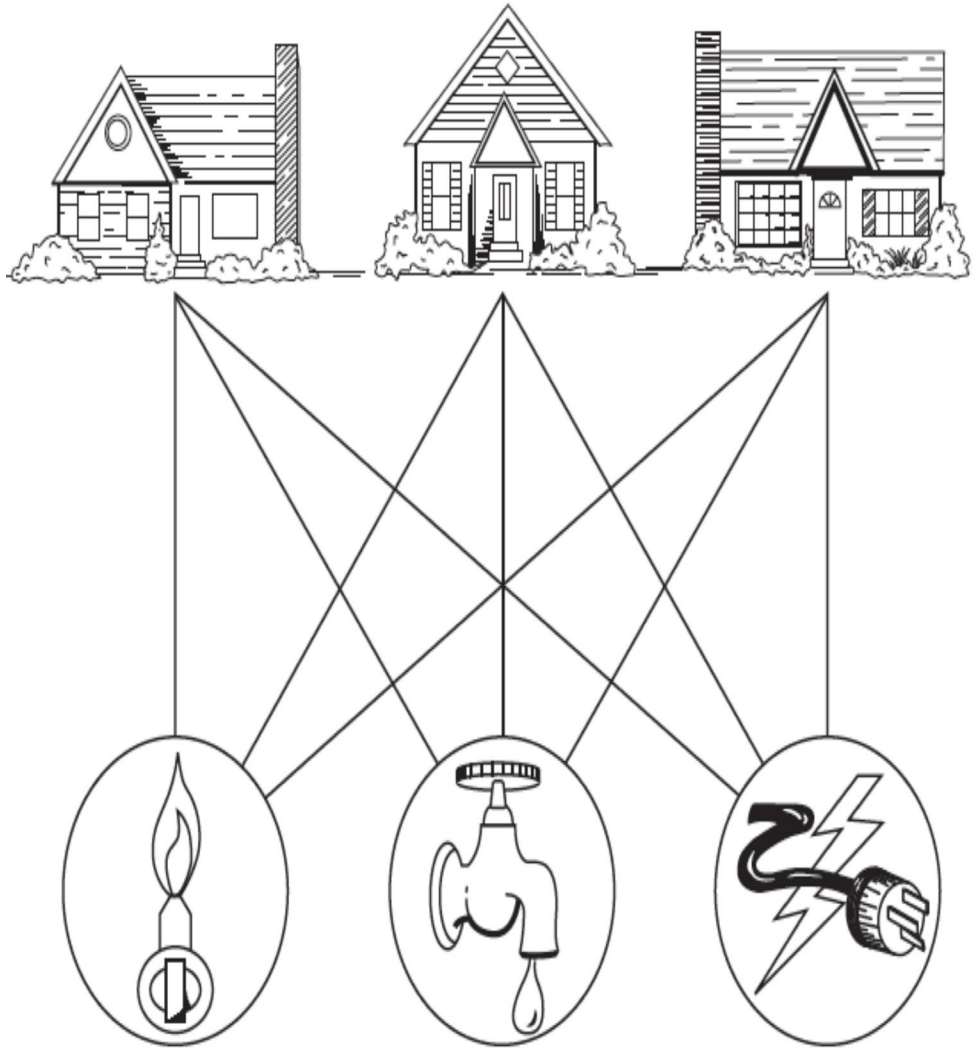
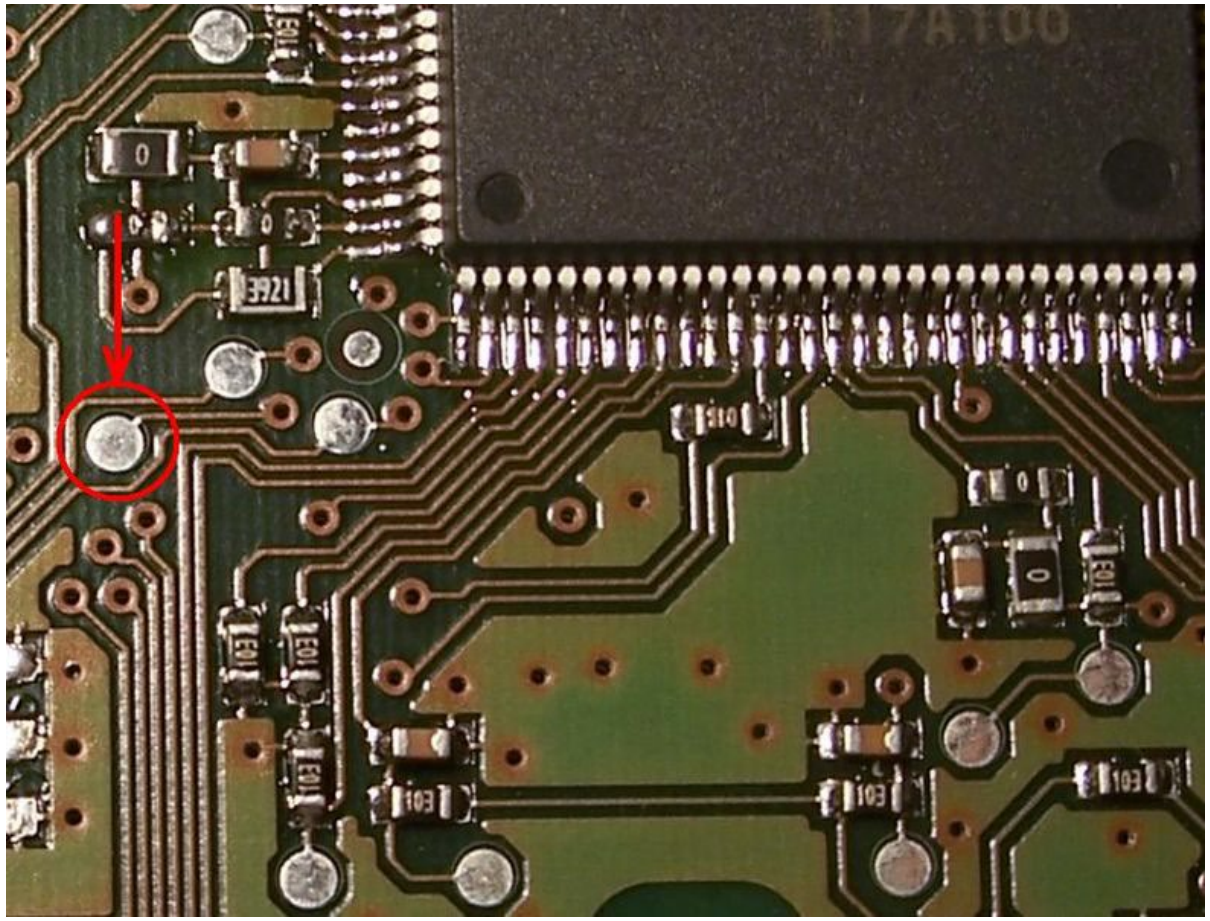
Planar Graphs

Planar Graphs





Planar Graphs

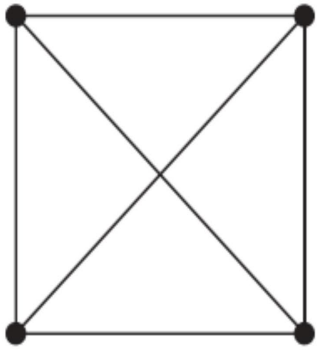
- A graph that can be drawn on a plane without edges crossing is called planar (e.g. a triangle)

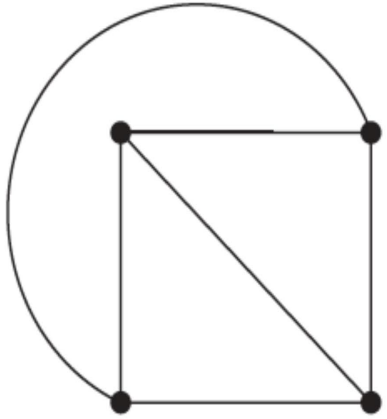
- Connecting utilities (electricity, water, natural gas) to houses. If we can keep from crossing those lines, it will be safer and easier to install

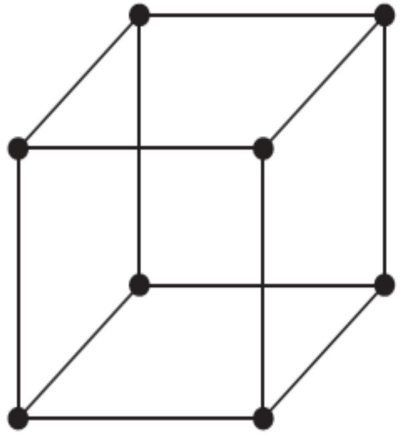
- Connecting components on a circuit board: the connections on a circuit board cannot cross. If we can connect them without resorting to another layer of traces, it will be cheaper to produce

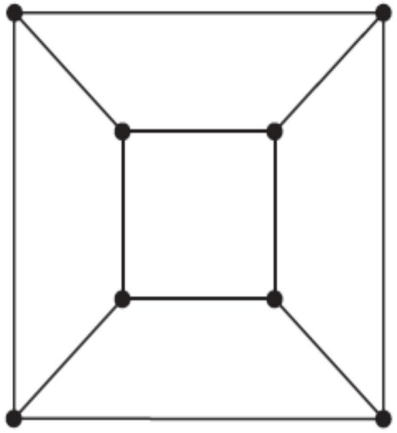
Planar Graphs

- The same graph can have many different drawings, so it is important not to confuse a particular drawing with the graph itself. In particular, planar graphs can have non-planar drawings



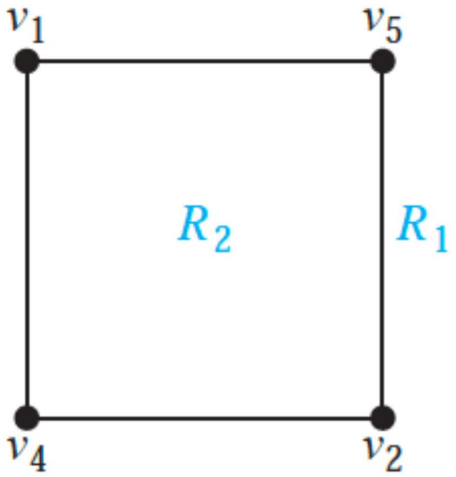


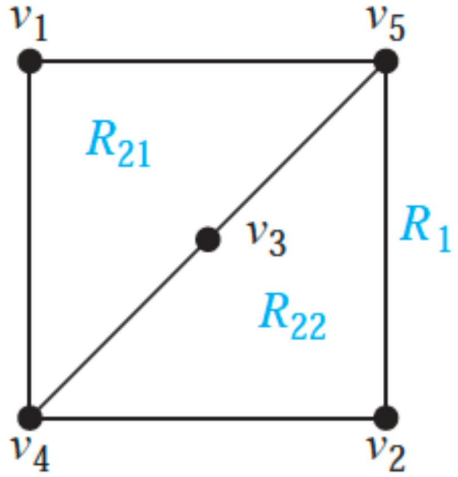




Plane Graph

- Plane Graph: planar graph that has been drawn in the plane without any edge crossings
- A plane graph divides the plane into regions (interior and exterior)





- Every plane graph has an unbounded region called the exterior region (Here, R1)

Faces

- In any planar graph, drawn with no intersections, the edges divide the planes into different regions
  - The regions enclosed by the planar graph are called interior faces of the graph
  - The region surrounding (outside) the planar graph is called the exterior face of the graph
  - When we say faces of the graph we mean BOTH the interior AND the exterior faces. We usually denote the number of faces of a planar graph by f

Faces



- The lefthand graph has three faces. The boundary of face 2 has edges df, fe, ec, cd, so this face has degree 4. The boundary of face 3 (the unbounded face) has edges bd, df, fe, ec, ca, ab. So face 3 has degree 6

Faces



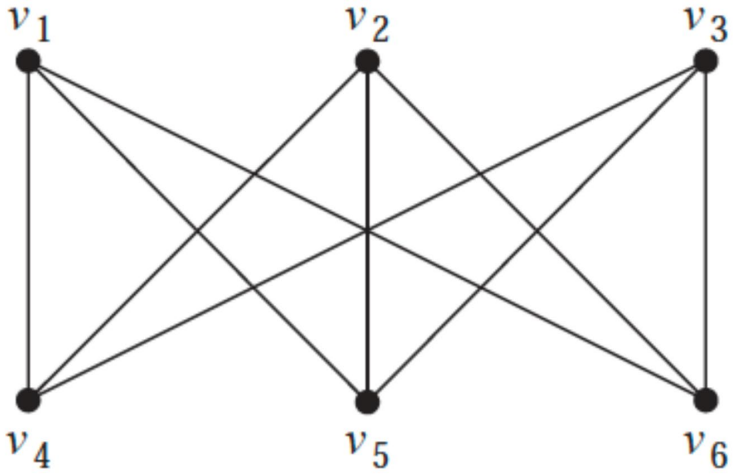
- The righthand graph above has a spike edge sticking into the middle of face 1. In such a case, the boundary must traverse the spike twice, i.e. the boundary of face 1 has edges bf, fe, ec, cd, cd, ca, ab, in which cd is used twice. So the degree of face 1 in the righthand graph is 7

Free Trees

- A free tree is any connected graph with no cycles. Free trees are somewhat like normal trees, but they don't have a designated root node and, therefore, they don't have a clear ancestor-descendent ordering

- A free tree doesn't divide the plane into multiple faces, because it doesn't contain any cycles. A free tree has only one face: the entire plane surrounding it

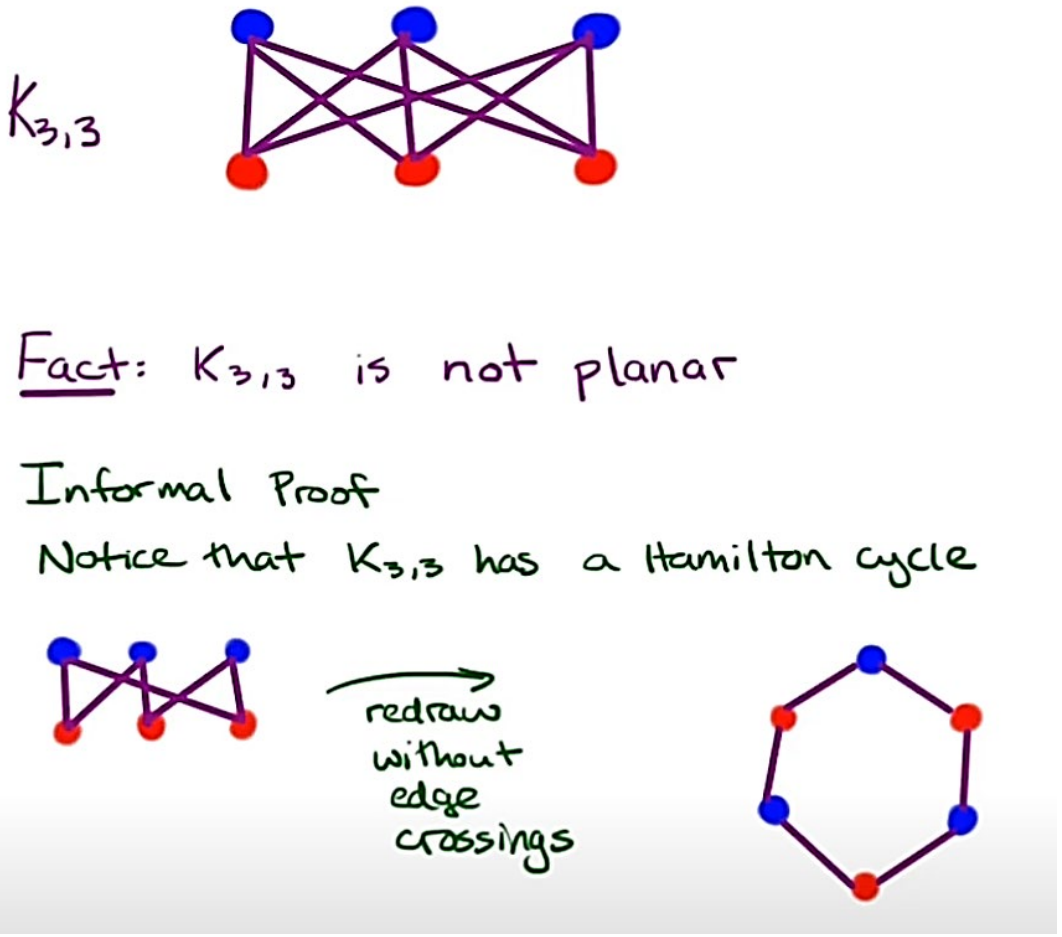
Example: Non-planar graph



- This is a $k_{3,3}$ graph
  - Complete bipartite graph with 3 vertices in the first set and 3 vertices in the second set

- Bipartite Graph
- A bipartite graph is a graph whose vertices can be divided into two disjoint sets, say U and V, such that every edge connects a vertex in U to a vertex in V
- No edges exist between vertices within the same set
- It is complete if every vertex in U is connected to every vertex in V

$K_{3,3}$ is not planar

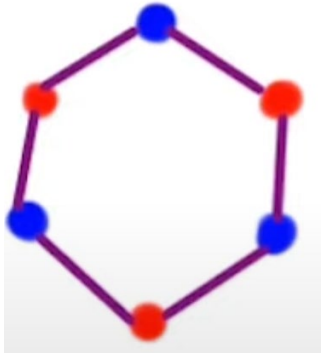


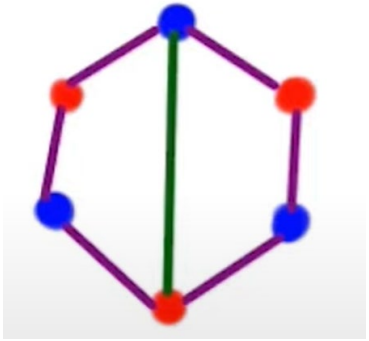
Hamilton Cycle: A cycle that visits each vertex exactly once

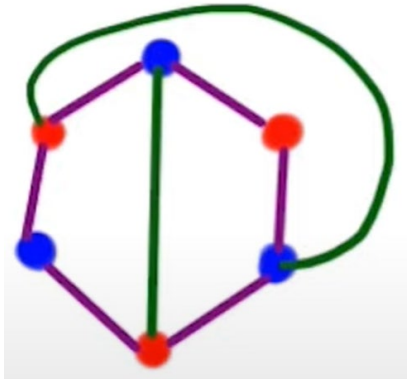
In the original graph, every blue vertex is connected to every red vertex

Each blue vertex is missing an edge to a red vertex

$K_{3,3}$ is not planar

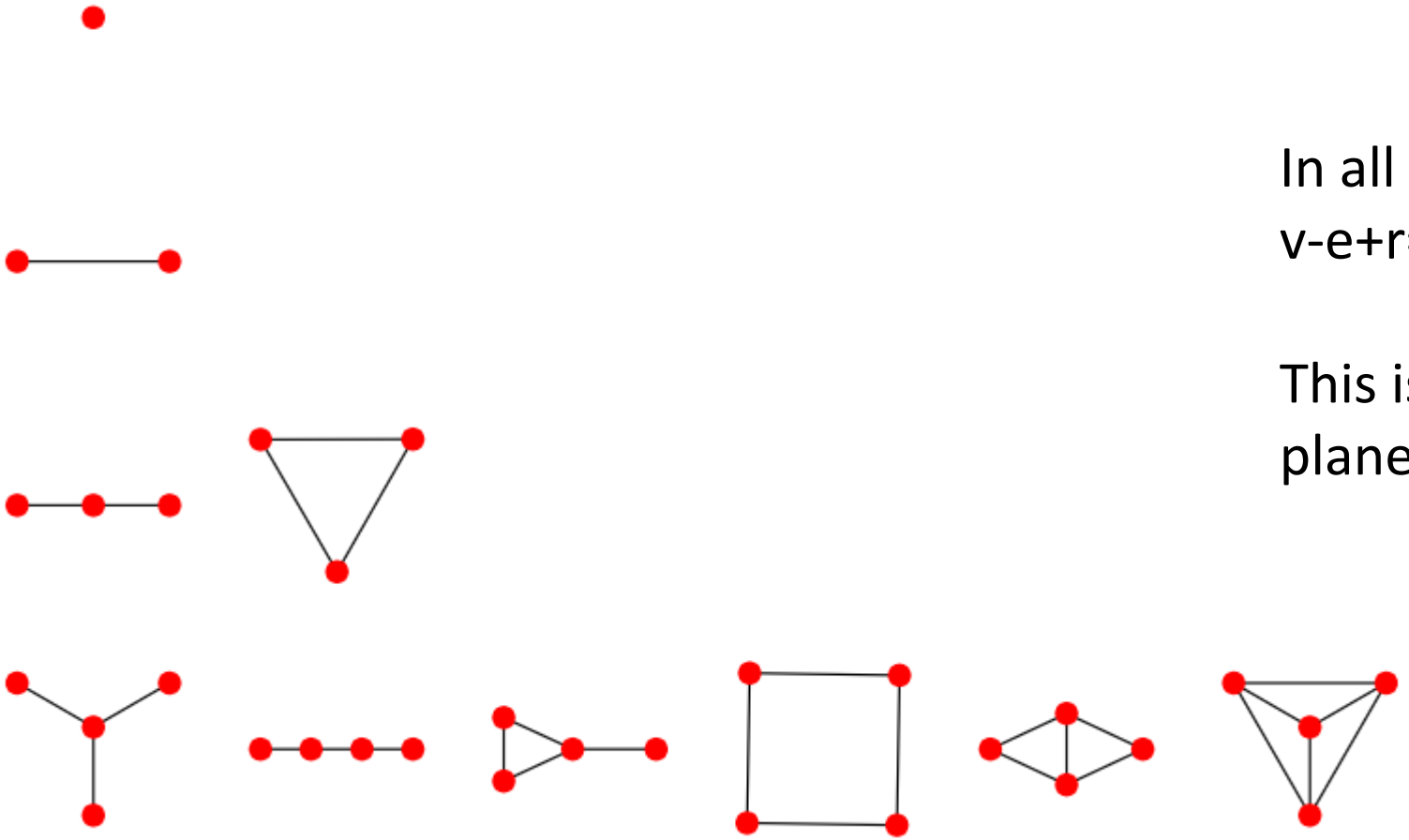






- We cannot draw the last edge without edge crossing

Connected Plane Graph



In all of these plane graphs:
v-e+r=2

This is the Euler's Formula for plane graph

Mathematical Induction

1. **Base Case**: Show that the statement is true for the smallest value (typically n=1 or n=0).
2. **Inductive Hypothesis**: Assume that the statement is true for some arbitrary n=k.
3. **Inductive Step**: Prove that if the statement is true for n=k, then it must also be true for n=k+1.
4. **Conclusion**: By induction, the statement holds for all $n \geq 1$.

Example

Prove that the sum of the first n positive integers is $S_n = n(n+1)/2$

Exercise

Prove that $2^n \geq n+1$ for all $n \geq 1$

Euler's Formula

- Let G be a connected planar simple graph with e edges and v vertices. Let r be the number of regions in a planar representation of G. Then v-e+r=2

Proof (induction on e)

- Base case: e=0
  - Since e=0, there is only one vertex and one region
  - v=1, e=0, r=1
  - Hence, v-e+r=2 holds

Euler's Formula

Inductive Hypothesis: Suppose the theorem is true for all connected plane graphs with fewer than 'e' edges (where e>=1)

- Now, consider a plane graph G with e edges, v vertices and r regions

- There are two cases we need to consider:
  - G is a tree and G is not a tree
- A graph is a tree if it is connected and acyclic

Euler's Formula

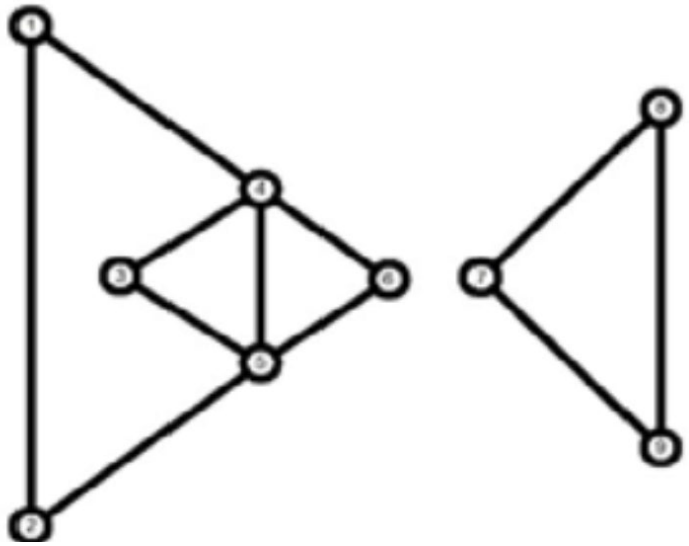
- If G is a tree
  - e = v-1
  - r = 1 (since it is a tree no region is closed off)
  - v-e+r = v – (v-1) + 1 = 2

- The statement holds

Euler's Formula

- If G is not a tree
  - Since G is connected, it has a cycle C
  - Let m be an edge of C
- We now want to remove 'm' and apply the inductive hypothesis in the smaller graph G\m (G without m)
  - There are v vertices, e-1 edges and r-1 regions
- So, v-e+r = 2 has to hold (hypothesis) for G\m
  - v-(e-1)+(r-1) = 2
  - v-e+r=2 (Holds)
- Therefore, the theorem holds for all m>=0

Corollary 1



If G is a plane graph with c(G) connected components

then: v-e+r = 1 + c(G)

Base Case: c(G)=1

Then, v-e+r=1+1=2 [This is the Euler's formula]

Inductive Hypothesis: Assume that for any plane graph with c(G)=k connected components, the formula holds: v−e+r=1+k

Corollary 1

Inductive Step c(G)=k+1: Consider a plane graph G with c(G)=k+1 connected components

Let G_1 be a subgraph consisting of k connected components, and G_2 be the additional connected component (disjoint from G_1)

By inductive hypothesis for G_1: $v_1-e_1+r_1=1+k$

For G_2: Since it is a single connected component: $v_2-e_2+r_2=2$

Corollary 1

Combine G_1 and G_2:

The total vertices, edges, and regions become: $v=v_1+v_2$,$e=e_1+e_2$,$r=r_1+r_2-1$. The -1 adjustment accounts for combining the regions from the disjoint components (adding one extra face for the outer region of G_2)

$$v-e+r = (v_1+v_2)-(e_1+e_2)+(r_1+r_2-1)$$

$$= (v_1-e_1+r_1)+(v_2-e_2+r_2)-1$$

$$= (1+k) +(2)-1$$

$$= 1+(k+1)$$

Thus, the formula holds for $c(G)=k+1$

Corollary 2

Suppose G is a connected planar graph, with v nodes, e edges, and f faces, where $v \geq 3$. Then $e \leq 3v - 6$

Proof:

In any planar graph, each face is bounded by at least 3 edges (this is true for $v \geq 3$, assuming there are no loops or multiple edges). However, since each edge is shared by at most two faces, the total number of edge appearances across all faces is 2e. Therefore, we have the inequality: $3f \leq 2e$

Corollary 2

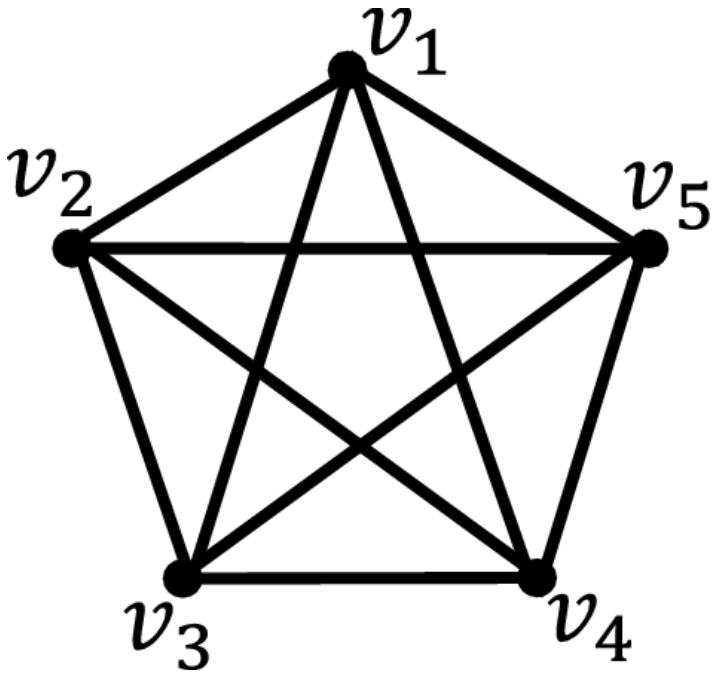
Rearranging for f: $f \leq 2e/3$

Substitute $f=e-v+2$ (from Euler's formula) into $f \leq 2e/3$:

$e-v+2 \leq 2e/3$

$e \leq 3v-6$

K_5 Graph

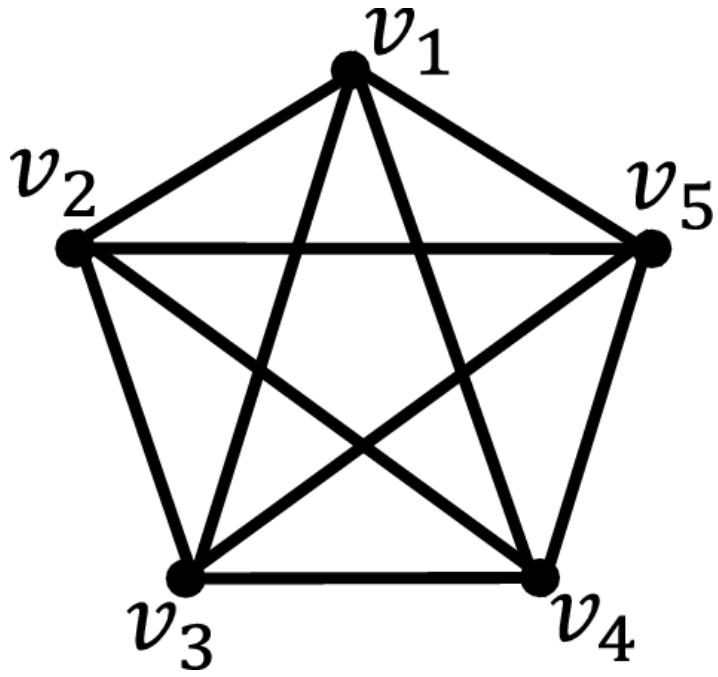


A **K_5 graph**, also known as the **complete graph on 5 vertices**, is a graph in which every pair of distinct vertices is connected by an edge

The total number of edges: e = v (v-1)/2

For K_5: e =10

K_5 Graph



For planar graph: v-e+f=2

Also, e≤3v−6 (for v≥3)

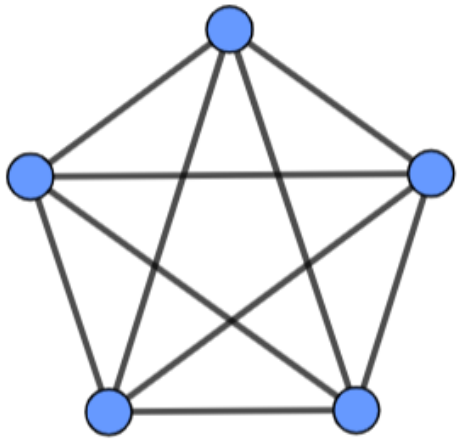
For K_5, 3v-6=9

But K_5 has 10 edges, not 9

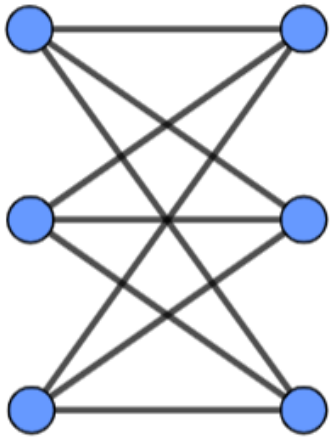
Hence it is non-planar

Kuratowski's Theorem

- A graph is planar if and only if it does not contain a subgraph that is a subdivision of K_5 or $K_{3,3}$



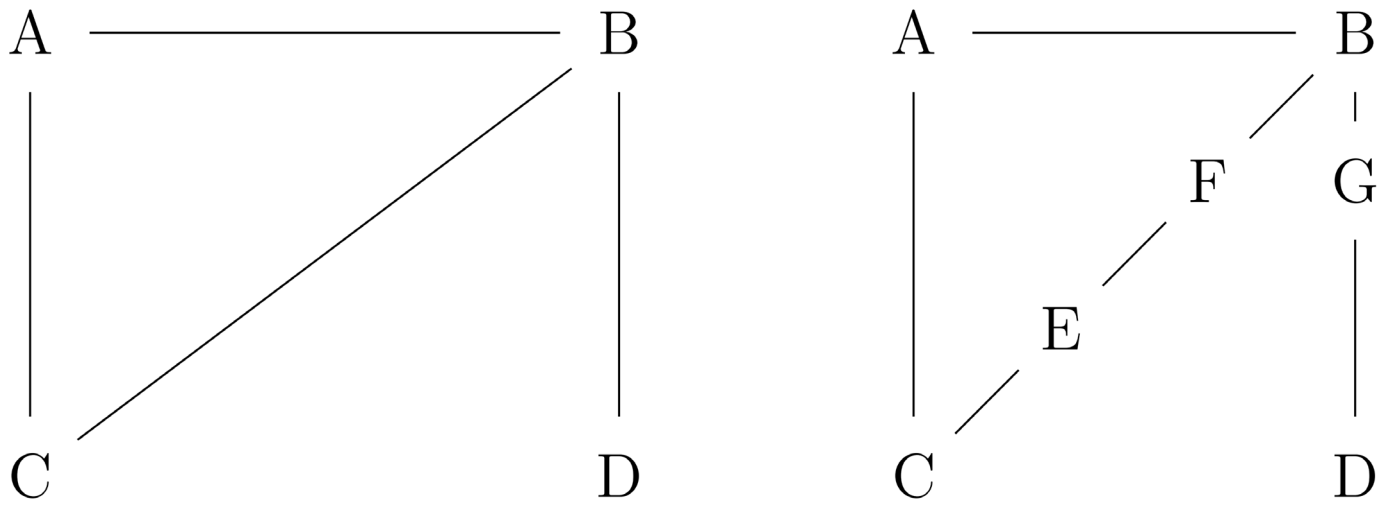
K_5



$K_{3,3}$

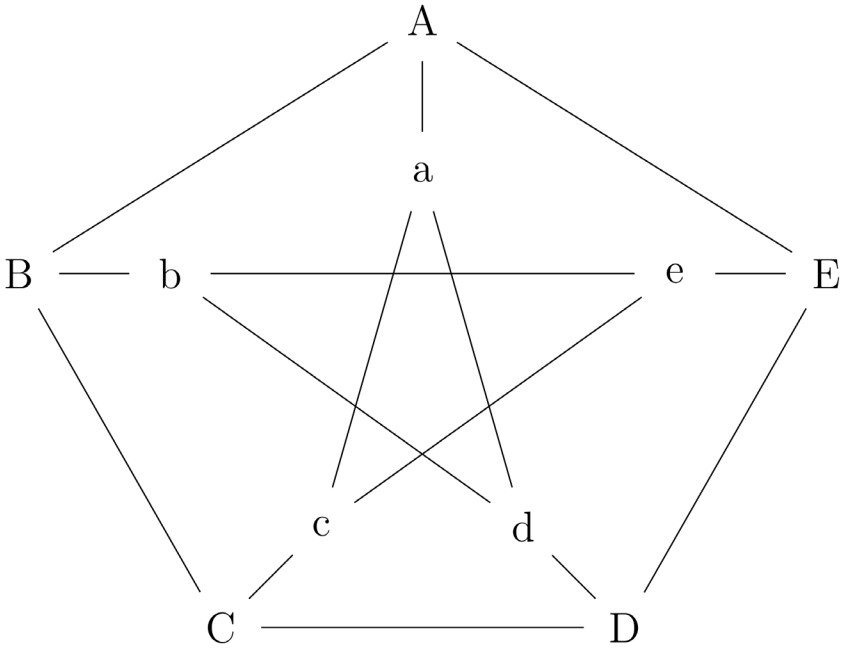
Subdivision

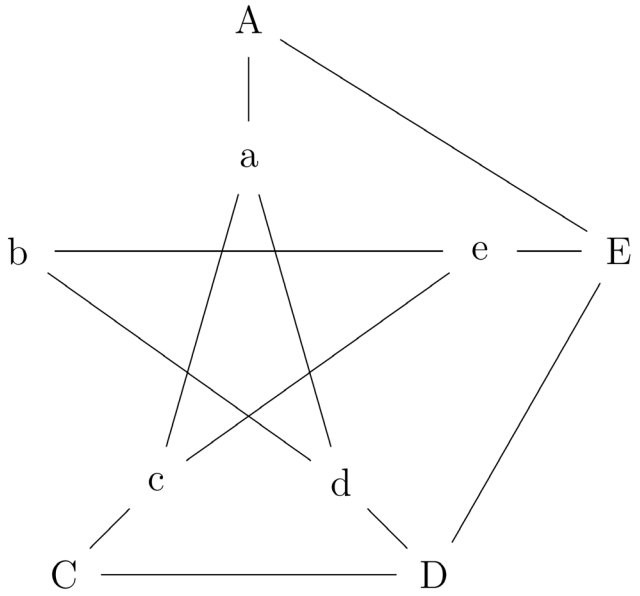
- A graph G is a subdivision of another graph F if the two graphs are isomorphic or if the only difference between the graphs is that G divides up some of F's edges by adding extra degree 2 vertices in the middle of the edges



Peterson's Graph

- v=10, e=15, degree=3, symmetric
- This is not planar
- Let us look at it's subgraph:

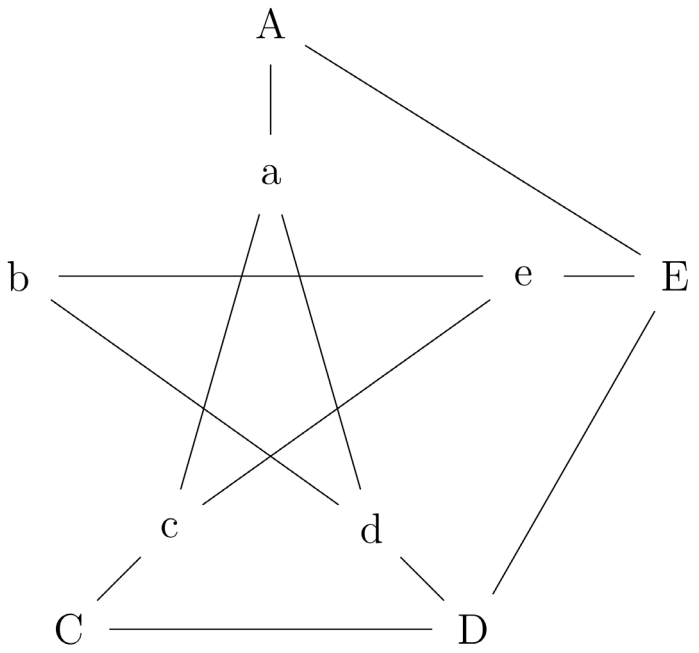




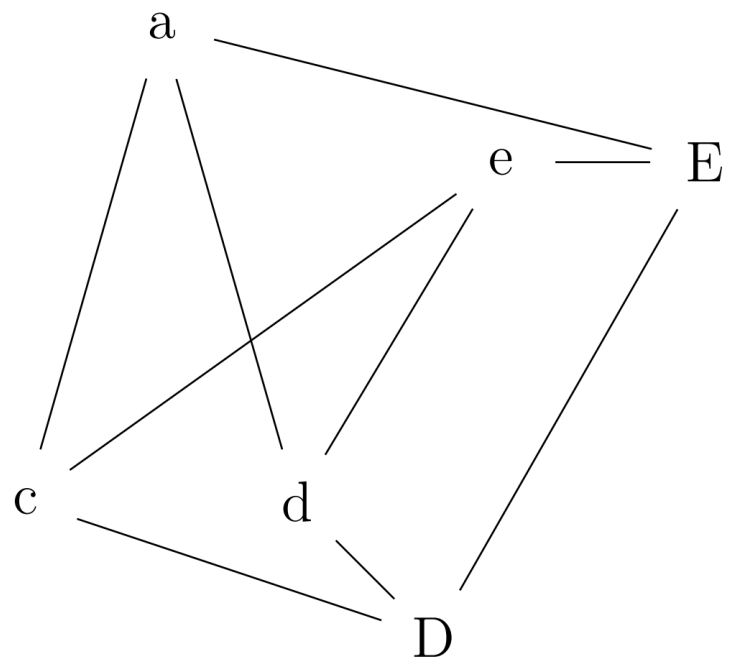
This subgraph is a subdivision of $K_{3,3}$

Peterson's Graph

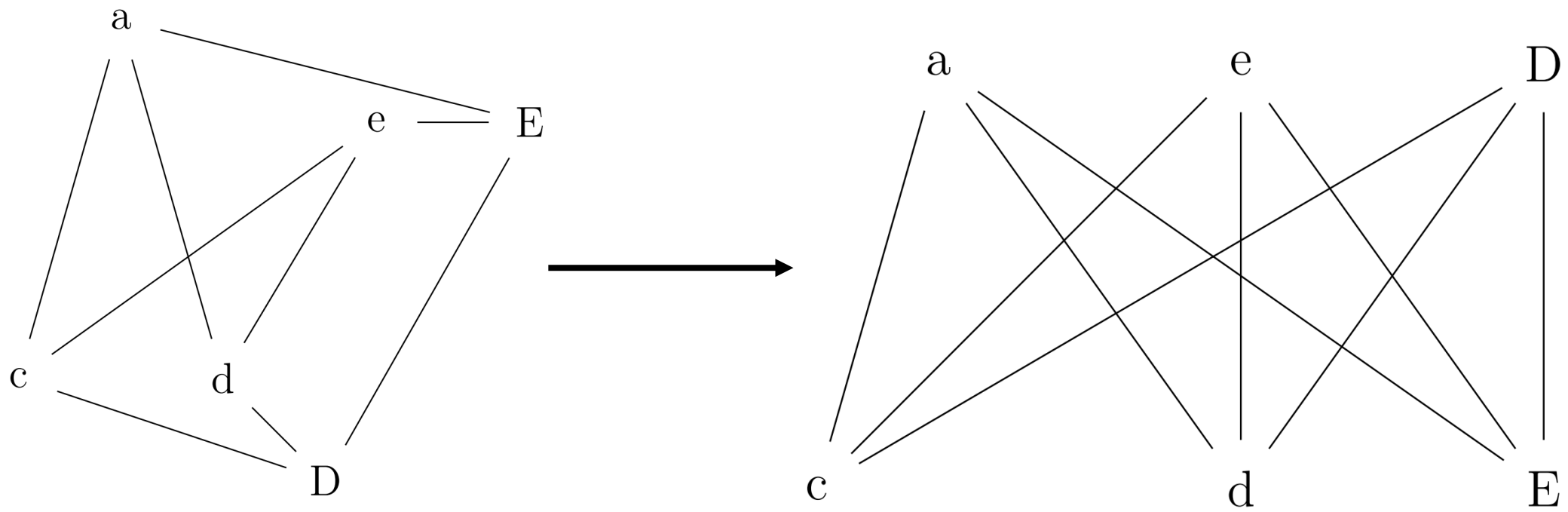
- Notice that node b is subdividing the edge from d to e. So we can delete it



- Similarly,
  - A is subdividing a and E
  - C is subdividing c and D



Peterson's Graph



This is a $K_{3,3}$ graph